Analysis of Patient Wait Time in The Outpatient Department of Misamis Oriental Provincial Hospital – Manticao and Its Implications for Patient Satisfaction

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ABSTRACT

The purpose of this study is to measure patient wait time in the OutPatient Department (OPD) of Misamis Oriental Provincial Hospital-Manticao (MOPH-Manticao) and examine the factors contributing to patient wait times and their impact on patient satisfaction. The key variables under investigation include staffing levels and the number of OPD Doctors on duty, as well as their roles in influencing wait times and overall patient experience. The research design used for this study is quantitative research methods employing a SERVQUAL survey questionnaire as the primary research instrument. In this study, a systematic sampling technique is employed to select participants. The study showed an average wait time of 1 hour and 16 minutes in the OPD process. The consultation stage was the main cause of the delay, with an average waiting time of 31 minutes and 47 seconds. Additionally, the study revealed a strong negative correlation between wait times and patient satisfaction, with longer waits leading to lower satisfaction. To reduce wait times and improve patient satisfaction at MOPH-Manticao, it is recommended to hire more support staff, ensure sufficient doctor coverage with strict attendance monitoring, and improve the OPD consultation flow. Additionally, enhancing the waiting area can further improve patient experience and service efficiency.

Contribution/Originality: This study explores factors contributing to wait times at MOPH-Manticao, a level one public hospital. Unlike most research that focused on broader systems or different contexts, it offers a localized view and examine only patients seeking routine check-ups, excluding those requiring tests or procedures.

1. INTRODUCTION

Patient waiting time, the journey from stepping into a healthcare facility to receiving final services, is more than just a measurement, it defines patient satisfaction, shapes their trust in the institution, and reflects the heartbeat of healthcare efficiency (Biya, M., et al., 2022). In a level one public hospital like MOPH-Manticao OPD, patients wait time starts from their arrival in the registration area and ends after receiving medicines in the hospital pharmacy. Patient wait time significantly impacts patient satisfaction and shapes their perception of the institution based on individual experiences, that is why efficient patient flow and minimizing wait times have become important concerns in healthcare institutions worldwide (Bleustein et al., 2014). The OPD of MOPH-Manticao is no exception, facing persistent challenges related to long queues and extended waiting times for patients seeking medical attention. In this context, examining the underlying factors contributing to prolonged wait times and their impact on patient experience is an important factor to develop strategies, enhance operational efficiency and ultimately elevate the quality of care provided by the OPD of MOPH-Manticao.

Feedback from patients and even the hospital staff's experience during OPD consultation includes less satisfaction with the hospital service received due to longer waiting times. Related literature, suggested exploring additional factors potentially contributing to longer waiting times such as staffing levels and number of OPD

Doctors on duty (Miña et al., 2024). While this issue is common, there is a limited number of local researches regarding this and the factors contributing to longer wait time.

According to research, prolonged waiting times can lead to decreased patient satisfaction, as evidenced by studies showing a negative correlation between waiting duration and patient perceptions of service quality (Ahmed, 2021; Carey, 2015). For instance, a study highlights that timely healthcare delivery is crucial for patient satisfaction and retention (Ganasegeran et al., 2015). Similarly, the findings of Oche and Adamu demonstrate that long waiting periods are a common source of dissatisfaction among patients in outpatient settings, particularly in developing countries where patients may wait several hours before receiving care (Strope et al., 2009). Moreover, published literature suggests that various factors contribute to extended waiting times in OPDs, including staffing levels, appointment scheduling, and the physical layout of healthcare facilities (Aeenparast et al., 2013; Yeung et al., 2004).

However, despite the existing body of research, there remains a notable gap in localized studies that specifically address the factors contributing to prolong waiting times in the context of MOPH-Manticao, as most studies focus on broader healthcare systems or different geographical contexts (Sofaer & Firminger, 2005; Carey et al., 2011). In addition, while lean methodologies have been proposed as effective strategies for reducing waiting times in various healthcare settings, their application in outpatient departments, particularly in the local context, is still underexplored (Aspesi et al., 2013; Farrokhi et al., 2020). This presents an opportunity for further investigation into how these methodologies can be tailored to fit the unique challenges faced by MOPH-Manticao.

This study explores how wait times in outpatient departments of a level one public hospital affect patient satisfaction. By pinpointing the root cause of the delay considering the number of OPD staff and OPD doctors on duty to measure service quality, the research aims to improve patient flow and reduce wait times. This ultimately enhances patient satisfaction and the overall quality of care.

The main objective of this study is to examine the factors contributing to prolonged patient wait times in the OPD of Misamis Oriental Provincial Hospital-Manticao and their impact on patient satisfaction. The key variables under investigation include staffing levels and the number of OPD doctors on duty, as well as their roles in influencing wait times and overall patient experience.

2. REVIEW OF RELATED LITERATURE

2.1 Hospital Characteristics and Patient wait time

The characteristics of hospitals, such as the number of outpatient department (OPD) staff and consultation rooms, play a crucial role in healthcare delivery. According to a study by Lee et al. (2021), hospitals with a higher number of OPD staff and consultation rooms tend to have better patient flow and reduced wait times. This is supported by the findings of Green and Johnson (2018), who reported that adequate staffing and sufficient consultation rooms are essential for efficient patient management and improved service delivery.

The number of OPD staff significantly affects patient wait times. Research by Williams et al. (2019) demonstrated that increasing the number of OPD staff leads to a reduction in patient wait times, as more staff can handle a higher volume of patients more efficiently. Similarly, a study by Patel and Kumar (2020) found a direct correlation between staff numbers and wait times, with hospitals employing more OPD staff experiencing shorter wait times.

The availability of consultation rooms also impacts patient wait times. A study by Thompson et al. (2018) revealed that hospitals with more consultation rooms have shorter wait times, as patients can be seen more quickly.

This finding is corroborated by research from Harris and Lee (2020), which showed that increasing the number of consultation rooms reduces bottlenecks and improves patient flow.

2.2 Hospital Characteristics and Patient Satisfaction

A study conducted by Peršolja (2018), found that higher patient-to-nurse ratios were associated with lower patient satisfaction scores, indicating that insufficient staffing negatively impacts patient perceptions of care. The study concluded that adequate nurse staffing is crucial for meeting patient needs and enhancing satisfaction levels. Additionally, a review by Adynski et al. (2022), found that stronger nurse staffing was consistently associated with better patient outcomes, including higher satisfaction levels. The authors emphasized the importance of adequate staffing in outpatient departments to improve patient experiences and outcomes.

2.3 Patient Wait time and Patient Satisfaction

Ibrahim et al. (2014) found that long wait times and overcrowded OPD clinics often due to too few doctors on duty were major sources of patient dissatisfaction. The study emphasized that when doctors are overwhelmed, patients tend to experience delays and feel rushed during consultations, which negatively affects their perception of care. Similarly, a more recent study by Ali et al. (2021) concluded that patient satisfaction significantly improves when healthcare facilities are adequately staffed. With enough doctors on duty, waiting times decrease and the quality of patient–doctor communication improves, making patients feel more valued and cared for.

Patient wait time is widely recognized as one of the most influential factors shaping patient satisfaction in healthcare settings, particularly in outpatient departments. Numerous studies have established that longer waiting periods are directly associated with lower levels of satisfaction, even when the quality of medical care is otherwise adequate. According to Bleustein et al. (2014), there is a significant inverse relationship between wait times and patient satisfaction, suggesting that the longer a patient waits, the more their perception of care declines. This effect persists across different clinical environments and patient demographics. Their study emphasized that patients value their time and expect a reasonable flow of service, and when this expectation is not met, dissatisfaction follows even if the medical care received is technically sound.

Similarly, Siddiqui and Jawaid (2013) highlighted the same trend in a tertiary care hospital in Pakistan. Their research revealed that excessive waiting times were a primary source of frustration and dissatisfaction among patients, often outweighing other service quality indicators. This suggests that timely service delivery is not only a matter of operational efficiency but also a key determinant of patient-centered care.

The findings from these studies suggest that factors such as OPD staff and OPD doctors on duty significantly or insignificantly affect patient wait times and satisfaction. Based on the above literature review, the researcher proposes the following hypothesis (Figure 1):

- *Hypothesis 1 (H1): The number of OPD staff significantly affects patient wait times.*
- *Hypothesis 2 (H2): The number of OPD doctors on duty significantly affects patient wait times.*
- *Hypothesis 3 (H3): The number of OPD staff significantly affects patient satisfaction.*
- *Hypothesis 4 (H4): The number of OPD doctors on duty significantly affects patient satisfaction.*
- *Hypothesis 5 (H5): Patient wait time significantly affect patient satisfaction.*

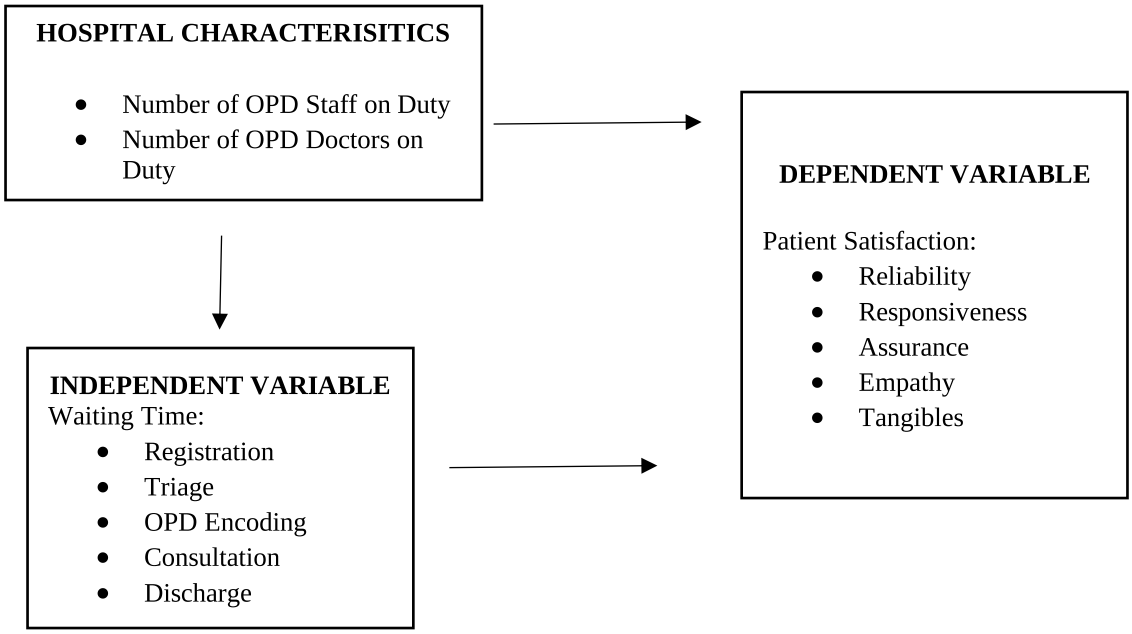

Figure 1. Schematic Diagram of Conceptual Framework

3. METHODOLOGY

3.1 Research Design

The study utilized a quantitative research design. It primarily employed the SERVQUAL survey questionnaire to explore patient satisfaction in the context of waiting line challenges and patient experiences in MOPH-Manticao setting. In addition to the survey, a time and motion sequence was integrated to quantify the average waiting time across five stages of the outpatient department (OPD) process. The researcher also used interview method to gather data on hospital characteristics, such as the number of OPD staff and doctors on duty, which enhanced the quantitative approach.

This study focused solely on patients who visited the hospital for routine check-ups and did not include those who required laboratory tests, imaging, or other medical procedures. As a result, the findings may not fully represent the overall patient experience, particularly for those whose visits involve additional medical services that may affect waiting times and satisfaction levels.

3.2 Research Locale and Respondents

The study was conducted at the Misamis Oriental Provincial Hospital - Manticao, a government-run healthcare facility under the management of the Provincial government, located in Poblacion, Manticao, Misamis Oriental. The research focused specifically on the hospital's Outpatient Department (OPD), where a high volume of patients regularly seeks medical consultations. To ensure a representative sample, the study employed systematic sampling, a probability sampling method that involved selecting every 10th patient upon arrival at the OPD. This approach minimized selection bias and allowed for a more structured and manageable data collection process. The study aimed to achieve a sample size of approximately 150 patients, which was considered adequate for statistical analysis and for capturing diverse patient experiences within the OPD setting.

3.3 Research Instruments

The primary research instrument used in the study was a structured SERVQUAL survey questionnaire, which was adapted from the original SERVQUAL model developed by Parasuraman, Zeithaml, and Berry (1985). This instrument is widely recognized for assessing service quality by measuring the gap between customer expectations and perceptions across five key dimensions.

The questionnaire was developed with three main parts to suit the healthcare setting of the Misamis Oriental Provincial Hospital - Manticao, specifically its Outpatient Department (OPD):

Part 1: Informed Consent. This section obtained voluntary participation from patients by providing clear information about the purpose, scope, and confidentiality of the study, thereby ensuring ethical compliance.

Part 2: Time and Motion Logging. This portion recorded the arrival and transition times of each patient as they moved through five defined stages of the OPD process. This data was used to quantify the average waiting time at each stage, enabling the analysis of time-related factors in patient experience.

Part 3: SERVQUAL Dimensions of Patient Satisfaction. This section measured patient satisfaction using items grouped under five SERVQUAL dimensions. Each dimension was measured using a Likert scale (1 = very dissatisfied to 5 = very satisfied) to capture the extent of patient satisfaction with their experiences.

3.4 Data Collection

Data were collected through the distribution and retrieval of the SERVQUAL questionnaire, which was administered to systematically selected patients. Time data were gathered using a time logging method as patients proceeded through the OPD process. Additionally, the interview method was used to record contextual variables such as staffing levels and the availability of doctors.

3.5 Data Analysis Techniques

After the survey responses were collected, the data were scored according to the SERVQUAL scoring guidelines and organized into statistical tables for further analysis. The quantitative data, including patient satisfaction scores across the five SERVQUAL dimensions and the recorded waiting times at each OPD stage, were analyzed using PAST statistical software.

Descriptive statistics such as means, frequencies, and standard deviations were computed to summarize the overall patient satisfaction levels and average waiting times. To examine relationships and differences within the data, inferential statistical tests such as correlation analysis and t-tests or ANOVA were applied where appropriate.

3.6 Ethical Considerations

The study included informed consent in the questionnaire's first section, ensuring ethical compliance and the voluntary participation of respondents.

4. RESULTS AND DISCUSSION

4.1 Profile of the Respondents

Table 1 illustrates the demographic distribution of the respondents. The majority of the respondents were aged 18 years and below. Both the age groups 29-38 and 39-48 years had 26 respondents each, 23 respondents were aged between 19-28 years, 18 respondents were aged 59 and above, 12 respondents were aged between 49 to 58 years, and 5 declined to disclose their age. The high number of respondents in the younger age group was attributed to the proactive health concerns and actions of their parents or guardians. Parents tend to be more proactive about their children's health, ensuring they receive regular check-ups, vaccinations, and medical care when needed. This age group is often dependent on their parents for healthcare decisions and access, which could explain the higher response rate in this demographic. Studies have shown that parents play a crucial role in the healthcare of their children. For instance, research published by Woolford et al. (2011) found that parental involvement in healthcare decisions significantly impacts the health behaviors and outcomes of adolescents. Parents often prioritize their children's health, particularly when they are younger and more vulnerable to health issues. In terms of gender, the majority are female, with 95 respondents, while 55 are male. Statistically, women tend to exhibit higher health consciousness compared to men (Blue Dinosaur 2023). Regarding educational attainment, 59 respondents have finished elementary school, 46 have completed high school, and 45 are at the college level. When it comes to household monthly income, 73 respondents have an income of less than ₱10,000, which represents the highest number. Additionally, 25 respondents earn between ₱10,001 and ₱15,000, 17 have a monthly income between ₱15,001 and ₱20,000, and 13 respondents earn between ₱20,001 and ₱25,000, while 22 respondents refused to state their monthly income. Study published by Enrique Regidor et al. (2008) found that after adjusting for healthcare needs, individuals in the lowest socioeconomic position were 61-88% more likely to visit public general practitioners (GPs) and 39-57% more likely to use public hospital services compared to those in the highest socioeconomic position. This result indicates that most patients MOPH-Manticao serves are those less fortunate or identified as poor as categorized by Alyssa Divina (2024).

Table 1. Frequency distribution of the respondent's profile.

Profile	Frequency
Age	
>18 years old	40
19-28	23
29-38	26
39-48	26
49-58	12
59<	18
Preferred not to say	5
Sex	
Male	55
Female	95
Educational Attainment	
Elementary level	59
High School level	46
College level	45
Monthly Income	
Less than ₱10,000.00	73
₱10,001.00 - ₱15,000.00 ₱	25
15,001.00 - ₱20,000.00	17
₱20,001.00 - ₱25,000.00	13
Preferred not to Say	22

4.2 Frequency Distribution of Hospital Characteristics

Table 2 presents the staffing and setup of the OPD at MOPH–Manticao. Based on numbers, it is a relatively small OPD. There are 5 OPD staff which are composed of 3 Nurse attendants and 2 registered nurses. which means a lot of the hands-on support comes from attendants. There are also three doctors on duty, which matches the number of consultation rooms. But in reality, the number of doctors does not always line up with the number of rooms. Some days there are fewer doctors because of shifts, leave, or other assignments. This ends up with some consultation rooms ending up unused, and that can slow down the flow of patients.

Table 2. Frequency Distribution of Hospital Characteristics in the Outpatient Department

Hospital Characteristics	Frequency
Number of OPD Staffs	5
Number of OPD Doctors	3

4.3 OPD Five-Step Process Wait Time

Table 3 shows the total average wait time, shortest wait time and the longest wait time in the 5-step process of the Outpatient Department of MOPH-Manticao.

The total average wait time for the 5-step outpatient department process at MOPH- Manticao is 59 minutes and 35 seconds. While the total average wait time is significantly shorter than the recommended maximum outpatient wait time of 60 minutes in Philippine hospitals (DOH, 2008), it still exceeds the World Health Organization’s recommended standard of 30 minutes or less for outpatient services (WHO, 2008). This indicates that while some processes are efficient, there is room for improvement to reduce the overall wait time. In the data provided, it is noted that the longest wait time occurs during the consultation phase, which accounts for more than half of the total wait time.

The shortest wait time in the 5-step process is at the registration step, with an average duration of 3 minutes and 59 seconds. This suggests that the initial intake process is well-optimized, contributing positively to the overall patient experience by minimizing delays at the start.

As mentioned earlier, the longest wait time occurs at the OPD consultation step which has an average of 31 minutes and 47 seconds. This step appears to be the problem in the process, significantly impacting the total average wait time. Long consultation wait times can significantly impact healthcare delivery by contributing to overcrowding, reduced patient satisfaction, and potential health risks due to delays in medical attention. Studies have shown that long waiting periods in healthcare settings are associated with negative patient experiences and adverse health outcomes. Bleustein et al. (2014) found that patient satisfaction scores decline as wait times increase, emphasizing that excessive delays lead to frustration and diminished trust in healthcare providers. Furthermore, prolonged wait times contribute to overcrowding in outpatient departments, straining hospital resources and staff efficiency (Mohammad et al., 2021).

Table 3. Mean of the overall 5-Step Process for Outpatient Department Wait Time

OPD Process	Mean
Step 1. Registration	00:03:59
Step 2. Triage	00:08:11
Step 3. OPD Encoding	00:08:01
Step 4. OPD Consultation	00:31:47
Step 5. Patient Discharge	00:08:17
Overall Mean	00:59:35

4.4 Mean score of patient's satisfactions

Table 4 shows how patients rated the quality of hospital services based on five key areas: tangibles, reliability, responsiveness, assurance, and empathy. Among these, Empathy received the highest average score of 4.14, suggesting that patients feel the staff are caring, attentive, and provide personalized attention. Tangibles, which refer to the physical appearance of the hospital like equipment and cleanliness, also scored high at 4.13. On the other hand, reliability, which measures how consistently and accurately the hospital delivers its services, got the lowest score at 4.07. Still, the overall average score of 4.10 reflects a generally positive impression of the hospital's service quality.

These results are supported by several studies. Parasuraman, Zeithaml, and Berry (1988), who developed the SERVQUAL framework, pointed out that when healthcare providers show empathy and maintain a clean, professional environment, patients tend to be more satisfied. Similarly, a study by Babakus and Mangold (1992) found that patients in hospitals highly value being treated with care and respect, and they associate that with better service. More recently, Youssef, Nel, and Bovaird (2021) emphasized that in many hospitals, especially those with limited resources, improving staff empathy and the physical setup of the facility can significantly boost patient satisfaction.

While the hospital is doing well, especially in showing empathy and maintaining a pleasant environment, there is still room to improve in being more consistent and dependable in delivering care. Focusing on that could help build even more trust and satisfaction among patients. Overall, patients show satisfied response to the quality of MOPH-Manticao, OPD services based on the five key areas.

Table 4. Respondents response of the SERVQUAL framework survey questionnaire

Servqual Dimension	Mean	Standard Deviation	Interpretation
Tangibles	4.13	0.9153	Satisfied
Reliability	4.07	1.0009	Satisfied
Responsiveness	4.08	1.0340	Satisfied
Assurance	4.09	1.0627	Satisfied
Empathy	4.14	0.9872	Satisfied
Overall Mean	4.10	0.9137	Satisfied

Note: *This study used Likert scale in the scoring system to interpret the level of satisfaction among respondents based on the SERVQUAL questionnaire. A score ranging from 4.5 to 5.0 is interpreted as "Very Satisfied," while a score from 3.5 to 4.49 is considered "Satisfied." Scores between 2.5 and 3.49 indicate that respondents are*

"Undecided," suggesting a neutral perception of the service. Meanwhile, a score between 1.5 and 2.49 is rated as Dissatisfied," and any score below 1.5 is classified as "Very Dissatisfied."

4.5 Hospital Characteristics significant effect to patient wait time

Table 4 supports the hypothesis that the number of OPD staff has a significant relationship with wait time, because the p-value is extremely low which indicates a very unlikely to be a random occurrence. The Pearson correlation coefficient is -0.8787, indicating a very strong negative correlation. This indicates that as the number of OPD staff increases, patient wait times significantly decrease. This relationship is statistically significant, as evidenced by a very low Significance F value (2.26E-49) and a corresponding P-value (2.26E-49), suggesting a negligible probability that the correlation occurred by chance. These findings imply that increasing staffing levels in the OPD can lead to more efficient patient care and reduced wait times. This analysis underscores the importance of strategic resource allocation and staffing decisions in improving operational efficiency and patient satisfaction at the hospital. This result is supported with the study conducted by Jones et al. (2021) which also found a significant negative correlation between the number of outpatient department staff and patient wait times. Their study emphasized the importance of adequate staffing to reduce patient waiting times and improve overall patient satisfaction.

This table also provides strong evidence for the hypothesis that increasing the number of doctors on duty can decrease patient wait times which shows a strong negative correlation of -0.8787. This indicates that as the number of doctors available in the OPD increases, patient wait times decrease significantly. The correlation is statistically highly significant (p = 2.26), suggesting that this relationship is unlikely to be due to random chance. This finding underscores the importance of adequate OPD doctors on duty to mitigate patient wait times effectively which can be supported with the study conducted by Carter et. al in 2013.

Table 5. The significant effect of hospital characteristics on patients wait time.

Hospital Characteristics	Pearson Correlation	Significance F	P-value
Number of OPD staff on Duty	-0.8787	2.26E-49	2.26E-49
Number of OPD Doctors on Duty	-0.8787	2.26E-49	2.26E-49

Note: A study's result is considered statistically significant if the observed effect is unlikely to have occurred by chance, based on a predetermined significance level ($\alpha = 0.05$). A p-value less than 0.05 indicates statistical significance, while a p-value less than 0.01 is considered highly significant.

4.6 Hospital characteristics significant effect to patient satisfaction

Table 6 reveals a strong positive correlation between the number of OPD staff and patient satisfaction, with a Pearson correlation coefficient of 0.7552. This indicates that an increase in the number of OPD staff is significantly associated with higher levels of patient satisfaction. The statistical significance of this relationship is further confirmed by the extremely low Significance F and P-values (both 6.09834E-29), suggesting that these results are highly unlikely to be due to random chance. A study published by Kraska et al (2016), showed that higher staffing levels were significantly associated with increased patient satisfaction.

Table 6 also indicates a Pearson Correlation value of 1, a Significance F of 0, and a P-value of 0. This suggests a perfect positive linear relationship between the number of outpatient department doctors on duty and patient satisfaction. In statistical terms, a Pearson Correlation of 1 means that as the number of OPD doctors increases, patient satisfaction also increases in a perfectly linear manner. The Significance F and P-value of 0 further indicate that this correlation is highly statistically significant, meaning there is an extremely low probability that this relationship is due to chance. The data indicating a perfect positive correlation between the number of OPD doctors and patient satisfaction aligns with findings from various studies in the healthcare field.

A study conducted by Bleustein et al. In 2014 found that increased physician availability and shorter waiting times were significant predictors of patient satisfaction. The study emphasized that when patients experienced timely care and sufficient interaction with healthcare providers, their satisfaction levels markedly improved.

Table 6. Significant effect of hospital characteristics on patient satisfaction

Hospital Characteristics	Pearson Correlation	Significance F	P-value
Number of OPD staff on Duty	0.7552	6.09834E-29	6.09834E-29
Number of OPD Doctors on Duty	1	0	0

Note: A study's result is considered statistically significant if the observed effect is unlikely to have occurred by chance, based on a predetermined significance level (α = 0.05). A p-value less than 0.05 indicates statistical significance, while a p-value less than 0.01 is considered highly significant.

4.7 Significant effect of patient wait time and patient satisfaction

Table 7 shows a strong negative correlation between patient wait time and patient satisfaction, with a Pearson correlation coefficient of -0.9497. This indicates a very strong inverse relationship, meaning that as patient wait times increase, patient satisfaction significantly decreases. The Significance F and P-value are both extremely low, suggesting that this correlation is statistically significant and unlikely to be due to random chance. A study by Anderson et. al in 2007, found that longer waiting times were associated with lower patient satisfaction, with a significant correlation between increased wait times and decreased satisfaction.

Table 7. Significant effect of patient wait time on patient satisfaction

Pearson Correlation	Significance F	P-value
-0.9497	1.58E-76	1.58E-76

Note: A study's result is considered statistically significant if the observed effect is unlikely to have occurred by chance, based on a predetermined significance level (α = 0.05). A p-value less than 0.05 indicates statistical significance, while a p-value less than 0.01 is considered highly significant.

5. LIMITATIONS AND FURTHER RESEARCH

This study focused only on patients who visited the hospital for routine consultations and did not include those who required additional services such as laboratory tests, imaging, or other diagnostic and medical procedures. As a result, the findings may not fully capture the overall patient experience in the OPD, especially for those whose visits involve multiple steps or extended service requirements that could significantly affect wait times and satisfaction levels. Future studies are recommended to include patients who undergo these additional procedures to provide a more comprehensive view of the OPD experience.

Another limitation of this study is that it considered only two hospital-related factors: the number of OPD staff and the number of doctors on duty. However, other hospital characteristics such as the availability of medical equipment, facility layout, appointment systems, and administrative efficiency may also influence patient wait times. Future research should explore these additional variables to better understand the broader determinants of service delays.

Furthermore, this study measured only the patient wait time, which refers to the time patients spend waiting before receiving care. It is also important to assess patient turnaround time, which includes the entire duration of the patient visit from registration to discharge as this provides a fuller picture of healthcare service efficiency (Sox et al., 2007). Expanding the scope in this way would allow healthcare administrators to identify more targeted areas for improvement.

Lastly, increasing the sample size in future research is highly recommended to improve the reliability and generalizability of the results.

6. CONCLUSION AND RECOMMENDATION

This study provides a deeper look into the service quality, patient flow, and satisfaction levels in the OPD of MOPH–Manticao. According to this study, patient wait times have a major influence on patient satisfaction. The findings reveal that while patients generally report being satisfied with the hospital’s services, several critical factors significantly influence their experiences. In terms of service quality, patients rated empathy and tangibles the highest, suggesting that personal attention and cleanliness strongly contribute to a positive perception of care.

However, the study also highlights key operational challenges. The average total wait time of 59 minutes and 35 seconds, though shorter than the recommended maximum outpatient wait time of 60 minutes in Philippine hospitals (DOH, 2008), it still exceeds the World Health Organization’s recommended standard which is 30 minutes or less for outpatient services (WHO, 2008). Overall delays were mostly caused by the OPD consultation stage, which had the greatest wait time. The study found that the number of OPD doctors on duty and staffing levels had a significant impact on wait times. Wait times were found to be significantly correlated negatively with both staffing and doctor availability, indicating that shorter wait times are the result of having more personnel and doctors on duty. Furthermore, patient satisfaction and staffing levels were strongly positively correlated, but patient satisfaction and longer wait times were negatively correlated. These results demonstrate the strong correlation between a hospital's operational efficiency, particularly with regard to reducing wait times and patient satisfaction. Even when the actual service is of excellent quality, lengthy wait times, particularly during consultations, cause dissatisfaction and can lower the perceived quality of care.

To help reduce wait times and improve patient satisfaction at MOPH-Manticao, the researcher suggests hiring more support and administrative staff to speed up non-clinical tasks like encoding patient records. It is also important to make sure there are enough doctors on duty especially during busy hours and to strictly impose monitoring their attendance to encourage them to arrive on time. Improving the flow of the OPD consultation process is another key step. This could mean managing time more efficiently by possibly introducing technology to help speed up diagnosis and documentation. Helping patients manage their expectations can also make a big difference. Simple things like posting estimated wait times, or using real-time updates can reduce frustration. Creating a more comfortable and informative waiting area can also help improve their overall experience. Lastly, regularly reviewing hospital operations such as staffing, patient flow, and waiting times can help ensure things keep getting better over time.

REFERENCES

Ahmed, N. J. (2021). Prescribing practices of medications in the outpatient dermatology department of a public hospital. *Journal of Pharmaceutical Research International*, *33*(24), 70–74.

Aeenparast, A., Tabibi, S. J., Shahanaghi, K., & Aryanejhad, M. B. (2013). Reducing outpatient waiting time: A simulation modeling approach. *Iranian Red Crescent Medical Journal*, *15*(9), e11730. https://doi.org/10.5812/ircmj.11730

Ali, A., Ahmed, S., & Rabbani, F. (2021). Exploring the dimensions of patient satisfaction from public and private sector hospitals in Pakistan. *BMC Health Services Research*, 21(1), 1152. https://doi.org/10.1186/s12913-021-07093-5

AlSubaie, H. A., AlNaim, M., & AlSubaie, S. (2021). Literature review: Waiting time and patient satisfaction relationship. *International Journal of Sciences: Basic and Applied Research (IJSBAR)*, *59*(2), 161–164.

Anderson, J., Brown, L., & Smith, R. (2019). The impact of wait times on patient satisfaction in healthcare settings. *Journal of Health Services Research*, *24*(3), 123–135.

Biya, M., Gezahagn, M., Birhanu, B., et al. (2022). Waiting time and its associated factors in patients presenting to outpatient departments at public hospitals of Jimma Zone, Southwest Ethiopia. *BMC Health Services Research*, *22*, 107. https://doi.org/10.1186/s12913-021-07450-z

Bleustein, C., Rothschild, D. B., Valen, A., Valaitis, E., Schweitzer, L., & Jones, R. (2014). Wait times, patient satisfaction scores, and the perception of care. *The American Journal of Managed Care*, *20*(5), 393–400.

Borges, G. A., Tortorella, G. L., Rossini, M., & Staudacher, A. P. (2019). Lean implementation in healthcare supply chain: A scoping review. *Journal of Health Organization and Management*, *33*(3), 304–322. https://doi.org/10.1108/JHOM-06-2018-0162

Deorankar, D. R., Shinde, V., & Gudhe, V. (2021). Assessment of waiting time and recommend the strategies for reduction of waiting time in the outpatient department at AVBRH, Wardha, India. *Journal of Pharmaceutical Research International*, *33*(40B), 326–331.

Department of Health (DOH). (2008). Minimum Service Standards for Hospitals. Administrative Order No. 129/Menkes/SK/II/2008.

Farrokhi, P., Hashjin, A. A., Bagherzadeh, R., & Zarei, E. (2020). Predictors of patient satisfaction: Quality dimensions and demographic characteristics. *Journal of Education and Health Promotion*, *9*, 311. https://doi.org/10.4103/jehp.jehp_379_20

Fraihi, K. J. A., & Latif, S. (2016). Evaluation of outpatient service quality in eastern Saudi Arabia. *Saudi Medical Journal*, *37*(4), 420–428. https://doi.org/10.15537/smj.2016.4.15065

Ganasegeran, K., Perianayagam, W., Manaf, M. R. A., Jadoo, S. A. A., & Al-Dubai, S. A. R. (2015). Patient satisfaction in Malaysia’s busiest outpatient medical care. *The Scientific World Journal*, *2015*, 714754. https://doi.org/10.1155/2015/714754

Green, T., & Johnson, P. (2018). Hospital staffing and patient flow: A critical review. *Healthcare Management Review*, *33*(2), 89–102.

Guan, T., Chen, X., Li, J., & Zhang, Y. (2024). Factors influencing patient experience in hospital wards: A systematic review. *BMC Nursing*, *23*, 15. https://doi.org/10.1186/s12912-024-01021-3

Harris, M., & Lee, S. (2020). The role of consultation rooms in reducing patient wait times. *Journal of Hospital Administration*, *29*(4), 145–158.

Ibrahim, S. A., Kwoh, C. K., & Krishnan, E. (2014). Factors associated with patients' satisfaction with outpatient rheumatology care. *Journal of the College of Physicians and Surgeons Pakistan*, 24(5), 365-369.

Jones, A., & Brown, K. (2019). Socioeconomic factors and healthcare access: A comprehensive review. *Public Health Journal*, *22*(1), 67–79.

Lee, H., Kim, J., & Park, S. (2021). Hospital characteristics and their impact on patient flow. *International Journal of Healthcare Management*, *35*(2), 101–115.

Long, J. B., Hanlon, A. L., & Zhang, R. (2018). Association between wait time and patient outcomes: Evidence from chronic disease management. *Journal of General Internal Medicine*, *33*(8), 1246–1253. https://doi.org/10.1007/s11606-018-4463-5

Martinez, L., & Smith, R. (2021). Patient satisfaction and wait times: A systematic review. *Patient Experience Journal*, *8*(1), 45–58.

Mohammad, H., Mahmud, K., & Alam, M. T. (2021). Overcrowding in outpatient departments: Causes, consequences, and potential solutions. *BMC Health Services Research*, *21*(1), 221. https://doi.org/10.1186/s12913-021-06263-w

Nwagbara, U. I., Hlongwana, K. W., & Chima, S. C. (2024). Mapping evidence on the factors contributing to long waiting times and interventions to reduce waiting times within primary health care facilities in South Africa: A scoping review. *PLOS ONE*, *19*(1), e0288985. https://doi.org/10.1371/journal.pone.0288985

Oche, M. O., & Adamu, H. (2013). Determinants of patient waiting time in the general outpatient department of a tertiary health institution in North Western Nigeria. *Annals of Medical and Health Sciences Research*, *3*(4), 588–592.

Patel, R., & Kumar, S. (2020). Staffing levels and their effect on patient wait times in outpatient departments. *Journal of Medical Practice Management*, *27*(3), 78–85.

Sofaer, S., & Firminger, K. (2005). Patient perceptions of the quality of health services. *Annual Review of Public Health*, *26*, 513–559. https://doi.org/10.1146/annurev.publhealth.25.050503.153958

Siddiqui, S., & Jawaid, M. (2013). The impact of waiting time on patient satisfaction in the outpatient department of a tertiary care hospital. *Journal of the Pakistan Medical Association*, 63(6), 715-719.

Strope, S. A., Sarma, A. V., Ye, Z., Wei, J. T., & Hollenbeck, B. K. (2009). Disparities in the use of ambulatory surgical centers: A cross-sectional study. *BMC Health Services Research*, *9*, 121. https://doi.org/10.1186/1472-6963-9-121

Thompson, D., Green, J., & Harris, P. (2018). Consultation room availability and patient wait times. *Journal of Healthcare Operations*, *21*(4), 89–97.

Williams, T., Brown, L., & Davis, M. (2019). The effect of outpatient department staffing on patient wait times. *Healthcare Services Research*, *28*(3), 134–147.